

DS_Day01_07

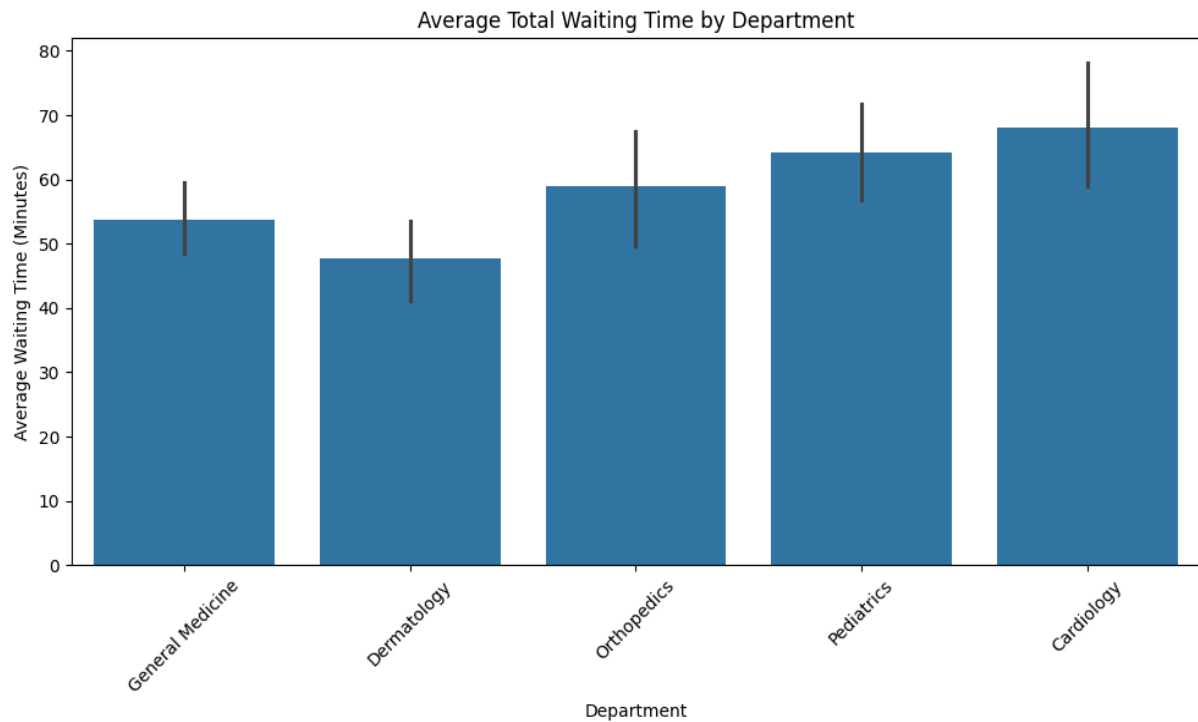
7. HOSPITAL — WHY ARE PATIENT WAITING TIMES HIGH?

key insights:

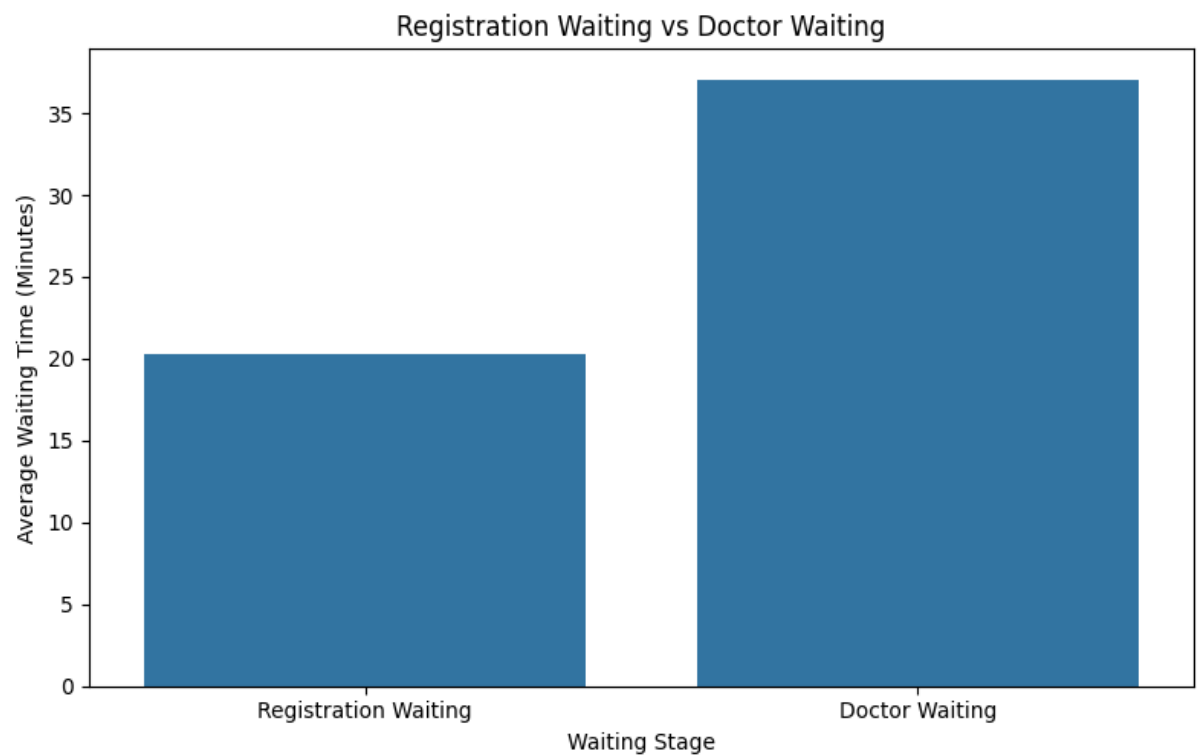
1. The average total patient waiting time is 57.4 minutes.
2. Doctor waiting time (37.07 minutes) is higher than registration waiting time (20.33 minutes), making doctor waiting the larger contributor to total waiting time.
3. Cardiology has the highest average waiting time at 68.13 minutes.
4. Dermatology has the lowest average waiting time at 47.63 minutes.
5. Pediatrics has the second-highest average waiting time at 64.11 minutes.
6. Appointments at 11 AM have the highest average waiting time (67.78 minutes), followed closely by 2 PM (67.07 minutes).
7. Weekday and weekend waiting times are relatively similar: 57.99 minutes on weekdays versus 55.89 minutes on weekends.

Visualisations

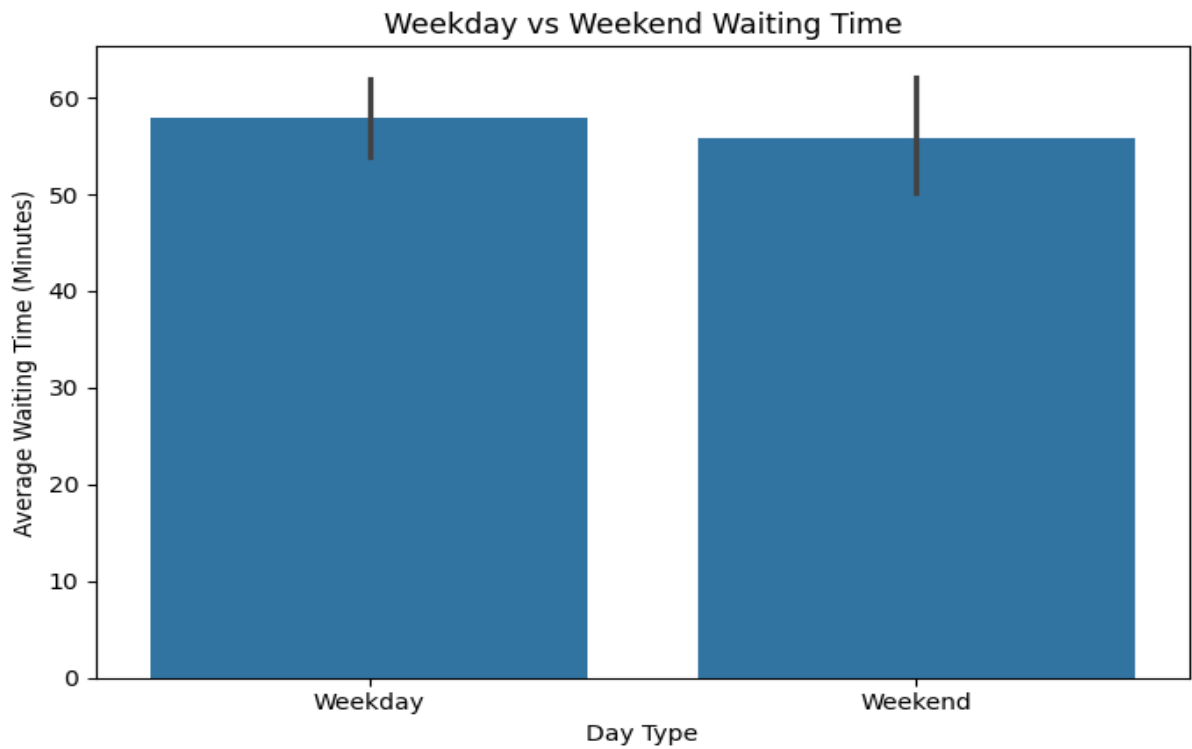
1. Average Total Waiting Time by Department



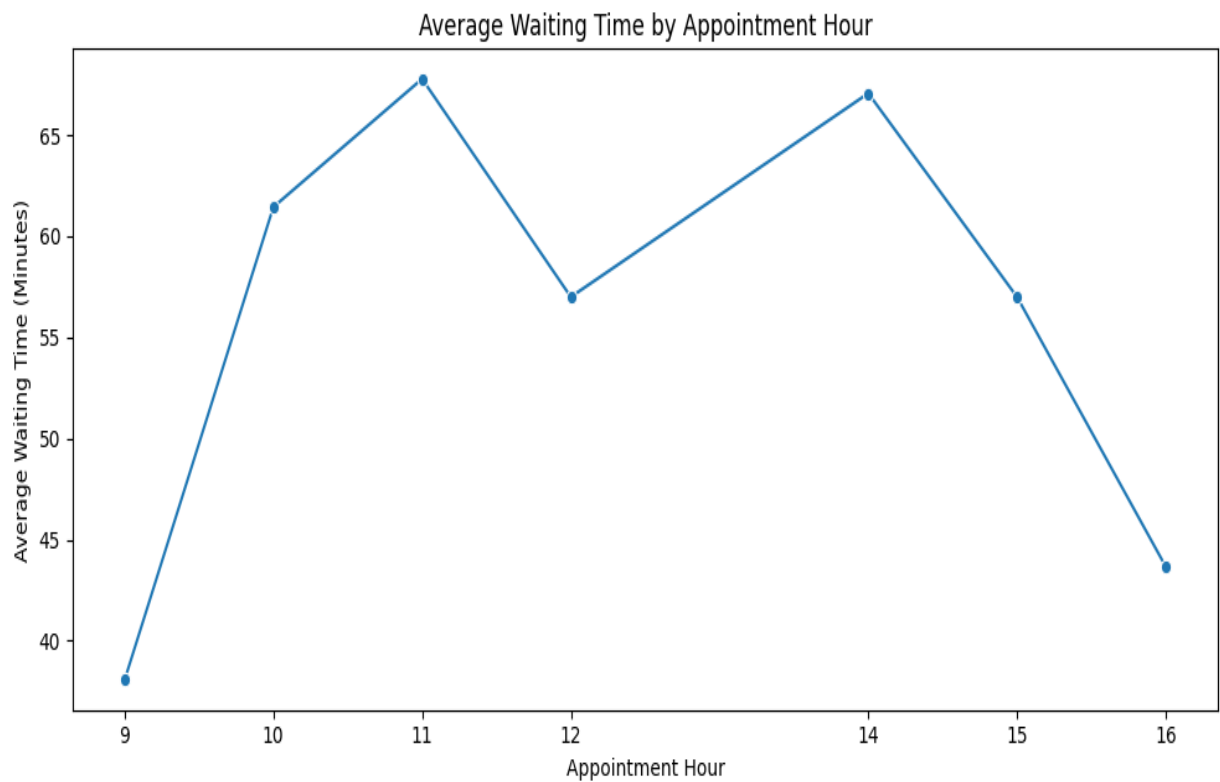
2. Registration Waiting vs Doctor Waiting



3. Weekday vs Weekend Waiting Time



4. Average Waiting Time by Appointment Hour



Model evaluation results:

Metric	Result
MAE	10.71 minutes
RMSE	13.41 minutes
R ² Score	0.108

Practical action plan.

1. Improve doctor availability during peak periods, especially around 11 AM and 2 PM, where waiting times are highest.
2. Review Cardiology operations, as it has the highest average waiting time of 68.13 minutes.
3. Spread appointments across available time slots to reduce congestion during high-waiting periods.
4. Improve the registration process, which contributes an average waiting time of 20.33 minutes.
5. Focus on reducing doctor waiting time, since it averages 37.07 minutes, higher than registration waiting time.
6. Continue monitoring weekday and weekend performance, as their average waiting times are relatively similar: 57.99 and 55.89 minutes, respectively.